

Multi-agent Large Language Model systems for Analyzing Elementary Students' Constructed Response

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Abstract

In this paper, we propose a multi-agent Large Language Model system to analyze elementary students' written responses to science tasks. The system analyzes student performances, identifies sources of uncertainty in the AI analysis process, and provides scores with scoring reasons as feedback for teachers and students. Applying our system to a dataset of elementary student responses, the system achieved a scoring accuracy of 90%, using human scores as the baseline. The strong alignment between AI-generated and human-assigned scores demonstrates the potential of the multi-agent system to enhance both the efficiency and quality of instruction. Furthermore, the model offers clear and consistent rationales that can support student learning and promote transparency in assessment.

1 Objectives

With the growing demand for more rigorous STEM education that fosters students' development of deep, usable knowledge, AI-based classroom assessments are becoming increasingly valuable (Knowledge, 2012). These assessments allow students to use and apply their knowledge to solve problems and receive timely feedback aligned with learning goals. Such AI-driven tools can analyze student performance objectively and provide personalized guidance for improving teaching and learning (Authors, 2024a). For feedback to be effective, it must involve analyzing student responses, evaluating performance, and identifying strengths and weaknesses. However, in practice, teachers often lack the time and resources necessary to deliver this level of feedback consistently, limiting its potential impact on student learning. Recent advances in automated assessment technologies, particularly those powered by Generative AI, offer promising solutions. These tools can streamline the feedback process by automatically scoring responses to open-ended tasks, reducing teacher workload while enhancing instructional support. Nevertheless, existing AI systems exhibit limitations within

educational contexts, particularly in elementary classrooms where students frequently produce spelling and grammatical errors and may struggle to articulate their ideas coherently. Furthermore, current AI systems lack comprehensive safeguards to detect and communicate potential inaccuracies in generated feedback with the teacher, thereby constraining their reliability and diminishing their potential to effectively support teaching and learning.

This study aims to develop a multi-agent system to identify uncertainties in student answers and pinpoint weaknesses in their performance on usable knowledge assessment tasks, offering actionable insights to support knowledge development in these specific areas. We applied our system to a fourth-grade elementary Earth Science assessment task with the following objectives: (1) analyze students' written responses; (2) provide meaningful interpretations of scoring decisions and estimate uncertainty; and (3) determine final scores and offer constructive feedback to help teaching and learning.

2. Theoretical Framework

Recently, with the rapid advancement of artificial intelligence (AI)—particularly following the emergence of Large Language Models (LLMs) such as generative pre-trained transformers—AI has begun to exert a growing influence on the field of education, offering the potential to meaningfully transform traditional assessment practices. These models can interpret and generate human-like text, enabling rich and insightful interactions between teachers and students.

Prompting techniques. Prompting techniques are inference-stage methods that aim to enhance the LLM performance by optimizing the input prompts. There are two crucial foundations for prompting techniques, *in-context learning capacity* and *chain-of-thought* (Wei et al., 2022). By *in-context learning capacity*, humans can directly embed new examples for the specific task in the prompt, avoiding the need to perform the computationally expensive finetuning method while humans can understand how they can modify the prompts. Studies have verified the effectiveness of this approach by providing examples for the tasks to allow the model to generate the output in a specific format. Moreover, the *chain-of-thought* (CoT) method uses examples with step-by-step reasoning to guide the LLM perform the intermediate steps. Various research observed that CoT outperforms the basic original *in-context learning*. Weak LLMs need more intermediate steps to understand the prompts and

generate reasonable results, while stronger ones may need fewer steps (Cui,2025). With CoT, LLMs can perform better with more complex tasks.

Our study deployed the prompt engineering techniques to instruct LLM to analyze students' answer, match the answers with the keys points in the rubrics then provide corresponding scores and reasoning. The CoT design will (1) separate the students' response into critical parts based on the rubrics, (2) verify logical connections between reasoning and alignments with our rubrics, (3) identify the possible incorrect statements, and (4) provide overall grading and the reasoning down to each rubric.

Multi-agent System. A Multi-Agent System involves multiple LLMs or LLM-based agents working collaboratively toward a shared objective. This decentralized approach is particularly well-suited for addressing complex tasks that exceed the capabilities of a single model (Guo et al., 2024). In the field of education, researchers have adopted multiagent frameworks to tackle diverse and challenging tasks in classroom settings. One such example, (Chu et al. 2024), is an automated grading system composed of three distinct agents: one responsible for scoring, another for monitoring and detecting errors in AI-generated responses, and a third for adapting the grading rubrics based on the feedback from the monitoring agent.

To enhance scoring accuracy and quantify the uncertainty associated with AI-generated analysis, we employ multiple LLMs guided by distinct prompt formulations. Each model generates a score independently, and the final score is determined through majority voting (Wang et al., 2022). Several key factors inform the design of these diverse prompts: (1) students may receive credit even if their response includes only intuitive or one-step reasoning toward the correct answer, rather than fully articulated explanations; and (2) the level of detail, emphasis, and number of examples included in the prompts can vary to influence the depth and consistency of the model's grading behavior.

3. Methods

Research question. This study investigates how a multi-agent LLM system analyzes student responses and identifies weaknesses in their written responses for usable knowledge tasks compared to a single-agent system. To investigate this question, we developed, analyzed, and compared the AI performance of a single-agent system and a multi-agent system using a polytomous dataset from 4th grade elementary school students.

To better understand the extent of students' usable knowledge, improve AI scoring accuracy, and refine expert-designed rubrics to better align learning goals with student performance, we conducted a three-phase experimental study. In Phase One, we randomly selected 30 from a pool of 327 student responses, each independently scored by three experts: two of whom authored the curriculum and assessment, and one with expertise in assessment and curriculum design principles. The experts achieved an inter-rater reliability of 0.87, after which they held a discussion to reach consensus and refine the rubric. This refinement involved incorporating more student responses as grading examples and restructuring the rubric for improved logical clarity. These adjustments helped the LLM better grasp the nuances and emphasis of the grading and reasoning tasks.

In Phase Two, we applied stratified sampling to select 30 additional, ensuring a balanced representation across score levels—particularly emphasizing low-score samples (0 or 1). The same three experts rescored these samples using the revised rubric from Phase One. Each response was scored independently to measure inter-rater disagreement and evaluate AI scoring reliability. The inter-rater reliability remained at 0.87, prompting further refinement of the rubric with more explicit clarifications for each grade level.

In Phase Three, to assess the alignment between expert and AI scores, we selected another 30 responses with varying AI-assigned uncertainty levels, 20 of which were flagged as uncertain, and had them independently scored by the experts. The inter-rater reliability in this phase is 0.90. The scores across all three phases (90 total responses) were used to evaluate the AI's performance. For the single-agent system, we calculated overall accuracy by comparing AI and human scores and recorded whether the AI scored higher or lower than the human consensus. For the multi-agent system, five independent agents generated scores for each response. A response was labeled as uncertain if fewer than four agents agreed on the same score. For example, a score pattern such as "1, 2, 1, 2, 1" would be flagged as uncertain, with the majority vote ("1") used as the final AI score.

4. Data Sources and Materials.

We used 90 written responses as described above, selected from 327 fourth-grade elementary students who completed an assessment task developed based on the Earth Science curriculum.

Assessment task and rubrics. The development of our assessment and rubrics are grounded in the three-dimensional (3D) learning outlined by the Next Generation Science Standards (NGSS)(States, 2013). This approach requires students to engage simultaneously in science and engineering practices, disciplinary core ideas, and crosscutting concepts to understand phenomena in STEM subjects and solve real-world problems (Authors, 2024b). Our assessment task is to evaluate students' ability to apply knowledge in novel but thematically related contexts. Specifically, students are asked to observe a phenomenon in which wind and water energy reshape the local landscape, and using knowledge from the class to explain how similar changes might occur over time in a different geographic location. We developed holistic rubrics to analyze student performance.

5. Results and Discussions

Table 2 presents the results for all 90 selected student samples. The overall scoring accuracy achieved by the single-agent system was 0.71, while the multi-agent system reached 0.91, representing a 20% improvement in accuracy through the use of multiple agents.

For the single-agent system, the accuracy in Phase One was 0.80 (5 incorrect out of 30), dropped to 0.63 in Phase Two (11 incorrect). The performance decline in Phase Two coincided with the introduction of more low-scoring examples, challenging the system's ability to evaluate ambiguous student responses. In Phase Three, which included 20 ambiguous samples and 10 other samples, the system continued to struggle, confirming its limitations in handling vague inputs.

In contrast, the multi-agent system, using a majority voting strategy, maintained a more stable performance across phases. It achieved an accuracy of 0.90 (3 incorrect out of 30) in Phase One, 0.87 (4 incorrect) in Phase Two, and 0.87 (4 incorrect) in Phase Three, resulting in an overall accuracy of 0.88 (11 incorrect out of 90). When incorporating uncertainty handling, where uncertain cases are referred to human experts and the remaining cases are

scored by AI, the accuracy further improved to 0.93, 0.90, and 0.90 for Phases One, Two, and Three respectively, and 0.91 overall.

These results indicate that the multi-agent system is more robust and consistent, particularly in evaluating responses with lower scores or ambiguous content. This suggests our multi-agent system is well-suited for analyzing elementary students' answers, which often feature grammatical errors, limited vocabulary, and occasional inconsistencies in logical flow. We are scoring the remaining 237 out of 327 responses using the final refined rubric and are analyzing the data to validate our multi-agent system and improve the accuracy with more insights from both AI and teachers. We will present the final analysis of 327 students' responses and techniques for improving AI accuracy, including prompts and chains of thought from both the single-agent system and the multi-agent systems at the AERA conference.

For future directions, given the limited availability of large-scale student assessment datasets (e.g., exceeding 10,000 samples), we plan to explore strategies to make LLM fine-tuning feasible. To support this, we may generate synthetic data and incorporate expert feedback to guide the fine-tuning process in a pedagogically meaningful way. Another direction is to integrate retrieval-augmented generation (Lewis et al., 2020) techniques, allowing the model to access and utilize large-scale external data sources. This will support the system in tailoring feedback more effectively to individual student needs and educational contexts.

6. Scholarly significance of the study

This study presents a multi-agent system designed to analyze elementary students' responses to a science assessment task. The system not only assigns scores but also generates explanations for the scoring and provides uncertainty estimates associated with the assigned scores. Our study highlights two strategies for improving AI scoring accuracy. First, iterative revisions to AI prompt indicate that including detailed contextual information about students' backgrounds, along with explicit rubric criteria and varied examples, enhances AI scoring performance. Second, incorporating teacher revisions of AI-

generated scores, particularly in cases where the AI identifies uncertainty, improve AI scoring accuracy. Our experiments suggest that a multi-agent system perform reliably on responses it is confident about, underscoring the value of human oversight in ambiguous cases. This supports the integration of a human-in-the-loop (HITL) technique, where human input is used to iteratively refine both the scoring process and output of the AI system, thereby improving scoring accuracy and enhancing the system's uncertainty quantification. Overall, this study shows the potential of LLMs to advance educational assessment. Specifically, Generative AI can meaningfully contribute to the automation and enhancement of classroom evaluation, potentially reshaping the future of in-class assessments.

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Figure 1. Assessment task and rubric

Now look at the photo below. Think about what you know about Earth features and time scales.  Learning Performance: Students will make observations of a landscape and identify at least two parts of the natural system. They will describe how these parts interact and explain how their interactions contribute to changes in the land over time. Students will use evidence from patterns in rock formations and the effects of weathering and erosion by water, ice, wind, or vegetation to support their explanations.	
6 (Q6) Look closely at the photo. Think about a system that has parts that work together. Identify two or three parts of a system and explain how they work together to cause change over time?	
Rubric: Grade 3 (Proficient) — Identifies two or more interconnected parts of the system; provides appropriate explanation of how parts interact; shows basic understanding of system change over time. Grade 2 (Developing) — Uses partial evidence/reasoning but lacks scientific terminology OR uses terminology but doesn't fully explain interactions. <i>Examples:</i> "Physical weathering and erosion work together to shrink the mountain, and it may take 200 years." "The water and trees, the snow the mountain the water helps trees grow bigger the can cause physical weathering." Grade 1 (Beginning) — Identifies at least one part of the system; shows minimal connection between parts; partial explanation even if not aligned with goals. <i>Examples:</i> "Rocks and water are my system. If the rocks fall onto the grass..."; "If the snow and rocks fall it can cause a flood."; "The mountains might get small in 100,000 years." Grade 0 (No Credit) — Lists parts without interactions OR makes incorrect claims. <i>Examples:</i> "Rocks, water, land." "Rocks and snow are my system. Snow melts the rocks." Grade 0 (No Credit) — Illegible, irrelevant, or no response. <i>Examples:</i> "I don't really think there is any energy." "no answer", "no".	

Table 2. Results of scoring accuracy, tendency, and uncertainty

Rubric	Human score	AI incorrect			AI uncertain
		Total incorrect	AI higher	AI lower	

		Single	Multi	Single	Multi	Single	Multi	Muti	Human
Phase One: 30 randomly selected responses (single-agent ¹ : .80; multi-agent system ² : .90, .93)									
0	0	-	-	-	-	-	-	-	-
1	1	0	0	0	0	0	0	0	0
2	19	2	0	0	0	2	0	5	2
3	10	3	2	0	0	3	2	2	2
Phase Two: 30 responses with a higher proportion of low scores (single-agent ¹ : .63; multi-agent system ² : .87, .90)									
0	2	2	1	1	1	1	0	0	1
1	10	3	0	3	0	0	0	1	1
2	4	0	0	0	0	0	0	0	1
3	14	6	2	0	0	6	2	2	1
Phase three: 30 responses with 20 uncertainty and 10 certain cases (single-agent ¹ : .63; multi-agent system ² : .87, .90)									
0	0	-	-	-	-	-	-	-	0
1	4	1	0	0	0	1	0	4	1
2	22	6	3	1	0	5	3	14	1
3	4	4	0	0	0	4	0	4	2
Total: 90 responses (single-agent: .71; multi-agent system ² : .88, .91)									
0	2	1	1	1	1	0	0	0	0
1	15	4	0	3	0	1	0	5	0
2	45	8	3	1	0	7	3	19	3
3	28	13	4	0	0	13	4	8	0

Note: 1. Single-agent model accuracy. 2. Multi-agent system accuracy: We report two multi-agent